In [4]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import warnings
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.cluster import KMeans
warnings.simplefilter(action="ignore")

dataset = pd.read_csv("/Users/apple/Downloads/Live.csv")

dataset.head()

dataset.tail()

model = KMeans(n_clusters=3)

model.fit(dataset)

labels = model.predict(dataset)

np.unique(labels, return_counts=True)

silhouettes = []
ks = list(range(2, 12))
for n_cluster in ks:
    kmeans = KMeans(n_clusters=n_cluster).fit(dataset)
    label = kmeans.labels_
    sil_coeff = silhouette_score(dataset, label, metric='euclidean')
    print("For n_clusters={}, The Silhouette Coefficient is {}".format(n_
    silhouettes.append(sil_coeff)

plt.figure(figsize=(12, 8))
plt.subplot(211)
plt.scatter(ks, silhouettes, marker='x', c='r')
plt.plot(ks, silhouettes)
plt.xlabel('k')
plt.ylabel('Silhouette score');

model = KMeans(n_clusters=4)

model.fit(dataset)

labels = model.predict(dataset)

np.unique(labels, return_counts=True)

dataset["labels"] = labels

dataset

group_zero = dataset[dataset["labels"]==0]["num_comments"].mean()

group_zero

group_one = dataset[dataset["labels"]==1]["num_comments"].mean()

group_one
```

```
group_two = dataset[dataset["labels"]==2]["num_comments"].mean()

group_two

group_three = dataset[dataset["labels"]==3]["num_comments"].mean()

group_three

group_zero = dataset[dataset["labels"]==0]["num_shares"].mean()

group_zero

group_one = dataset[dataset["labels"]==1]["num_shares"].mean()

group_one

group_two = dataset[dataset["labels"]==2]["num_shares"].mean()

group_two

group_three = dataset[dataset["labels"]==3]["num_shares"].mean()

group_three

status_type = dataset[["status_type_photo", "status_type_video", "status_

dataset = pd.concat([dataset["labels"],status_type.rename("status_type")]

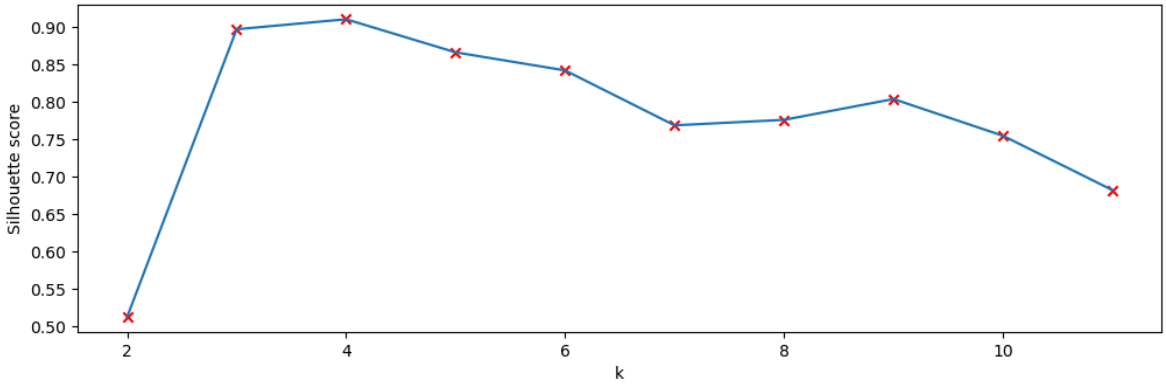
dataset.groupby(["labels","status_type"])["status_type"].count()
```

```
For n_clusters=2, The Silhouette Coefficient is 0.5121206343376229
For n_clusters=3, The Silhouette Coefficient is 0.8972493682244894
For n_clusters=4, The Silhouette Coefficient is 0.9106664962327098
For n_clusters=5, The Silhouette Coefficient is 0.86635373570091
For n_clusters=6, The Silhouette Coefficient is 0.8421404750401724
For n_clusters=7, The Silhouette Coefficient is 0.7685784340406729
For n_clusters=8, The Silhouette Coefficient is 0.7759265785299961
For n_clusters=9, The Silhouette Coefficient is 0.8038024216461908
For n_clusters=10, The Silhouette Coefficient is 0.7546219849351814
For n_clusters=11, The Silhouette Coefficient is 0.6818526335870103
```

Out[4]:

	labels	status_type
0	status_type_photo	4288
1	status_type_video	2334
2	status_type_status	365
3	status_type_photo	63

Name: status_type, dtype: int64



In [ ]: